Traceable Hybrid Auditing of Clinical LLM Outputs: Combining Deterministic Span Verification, Protocol Rules, and Graph-Based Reasoning Kernels

Hammam Omer
Faculty of Medicine, Omdurman Islamic University
Omdurman, Sudan
`Code & Benchmark: https://github.com/HammamNM/reasoning-debugger`

October 2026

Abstract

Evaluating Large Language Models (LLMs) in high-stakes clinical workflows such as acute chest pain management requires moving beyond final-answer accuracy toward rigorous process auditing. Current evaluation paradigms predominantly rely on unconstrained *LLM-as-a-Judge* frameworks, which are susceptible to judge hallucinations, unverified intermediate assumptions, and high false-positive rates when models express appropriate diagnostic uncertainty. In this paper, we present a **Traceable Neuro-Symbolic Auditing Framework** composed of three decoupled layers: (1) a `TraceExtractor` equipped with a deterministic substring verification firewall and canonical vocabulary normalizer, (2) a `ClinicalRuleEvaluator` enforcing temporal ordering, mandatory actions, and contraindications, and (3) a structural `Reasoning Kernel` that constructs Directed Acyclic Graphs (DAGs) to detect unsupported definitive inferences (`Pattern RP-001`). In a controlled validation and ablation study across synthetic adversarial traces, real frontier model outputs, and unseen linguistic edge cases, incorporating explicit assertion-status calibration (`PROPOSED`, `REPORTED_PERFORMED`, `NEGATED`, `SUSPECTED`, `DEFINITIVE`) increased auditor specificity from **57.1%** to **100.0%** and improved the overall F_1-score from **0.667** to **1.000** while maintaining **100.0%** sensitivity to protocol violations.

1 Introduction

As Large Language Models (LLMs) transition from static medical multiple-choice benchmarks (e.g., USMLE) [2] to open-ended clinical decision support, evaluating their procedural safety becomes critical. In emergency triage of **Acute Chest Pain** [1], an LLM may list a plausible differential diagnosis while simultaneously proposing unsafe management steps, such as discharging a patient after symptomatic relief from a gastrointestinal cocktail before serial troponin results are reviewed (premature closure), or initiating empiric anticoagulation in suspected acute aortic dissection.

Unconstrained *LLM-as-a-Judge* paradigms [3] exhibit two primary methodological vulnerabilities in clinical auditing:

1. **Extractor/Judge Hallucination:** The evaluating model may infer or fabricate clinical actions that were never stated in the candidate response.
2. **Epistemic Penalization:** Evaluators frequently conflate appropriate diagnostic uncertainty (e.g., listing NSTEMI as a *suspected* differential diagnosis) with an unsupported *definitive* claim, or misclassify *negated* warnings (e.g., *"Do not administer heparin"*) as proposed actions.

To address these gaps, we introduce a traceable, three-tier neuro-symbolic auditing pipeline that separates linguistic event extraction from deterministic rule and graph verification.

2 System Architecture

Our framework decomposes clinical auditing into three auditable stages:

2.1 Tier 1: `TraceExtractor` (Hallucination Firewall & Calibration)

Given a raw model response T, candidate clinical events $E = \{e_1, e_2, \ldots, e_n\}$ and support relations R are extracted and subjected to three deterministic filters:

1. **Deterministic Span Verification:** Every event e_i must provide a verbatim substring s_i. Let $\mathcal{N}(\cdot)$ denote whitespace and case normalization. An event is accepted if and only if:
$$\mathcal{N}(s_i) \subseteq \mathcal{N}(T) \tag{1}$$
Events failing this check are rejected with reason `SPAN_NOT_IN_SOURCE`.
2. **Canonical Vocabulary Normalization:** Lexical variants (e.g., `PERFORM_ECG`, `OBTAIN_ECG`) are mapped to canonical protocol codes (`ORDER_ECG`).
3. **Epistemic Assertion Calibration:** Each event is assigned an assertion tag $\alpha(e_i) \in \{$`PROPOSED`, `REPORTED_PERFORMED`, `OBSERVED_RESULT`, `NEGATED`, `SUSPECTED`, `DEFINITIVE`$\}$. Events with $\alpha(e_i) =$ `NEGATED` are excluded from active execution checks, and diagnostic nodes require structural graph proof if and only if $\alpha(e_i) =$ `DEFINITIVE`.

2.2 Tier 2: `ClinicalRuleEvaluator` (Domain Protocol Engine)

Verified traces are audited against structured clinical constraints:

- **Mandatory Omissions (`CRITICAL_OMISSION`):** Missing required actions (`must_include`) or diagnostic alternatives (`must_include_any`).
- **Temporal Ordering (`PREMATURE_CLOSURE`):** For any prerequisite constraint ($a_{\text{pre}} \prec a_{\text{target}}$), proposing a_{target} (e.g., `DISCHARGE`) when a_{pre} (e.g., `TROPONIN_RESULT_AVAILABLE`) is absent or occurs at $\text{idx}(a_{\text{pre}}) > \text{idx}(a_{\text{target}})$ triggers a violation.
- **Contraindications (`CONTRAINDICATED_ACTION`):** Proposing any action in the scenario's `must_avoid` set (e.g., `ADMINISTER_HEPARIN` in aortic dissection).

2.3 Tier 3: `Reasoning Kernel` (Structural Graph Verification)

Independent of domain rules, the `Reasoning Kernel` builds a Directed Acyclic Graph $G = (V, E_{\text{graph}})$ where edges represent evidentiary support (`EdgeRelation.SUPPORTS`). The `UnsupportedInferenceD` flags structural violation `RP-001` via:

$$\forall v \in V_{\text{INFERENCE}} : \quad \text{support_required}(v) = \text{True} \wedge \deg^{-}_{\text{SUPPORTS}}(v) = 0 \implies \texttt{RP-001} \tag{2}$$

3 Experimental Validation & Ablation Study

We validated the auditor across $N = 10$ benchmark traces (5 synthetic adversarial traces and 5 real frontier LLM responses) plus an unseen 5-case linguistic stress suite (`run_edge_cases.py`).

As shown in Table 1, removing assertion-status calibration caused 3 false-positive violations on safe responses that refused premature discharge (`NEGATED`) or framed NSTEMI as a differential possibility (`SUSPECTED`). Full calibration restored 100.0% specificity ($F_1 = 1.000$) and achieved 5/5 (100.0%) accuracy on the unseen edge-case suite.

Table 1: Ablation Study: Auditor Validation Against Human Gold Standard ($N = 10$)

Auditor Configuration	**TP**	**FP**	**TN**	**FN**	**Recall**	**Specificity**	F_1**-Score**
1. Baseline (No Assertion Filt.)	3	3	4	0	100.0%	57.1%	0.667
2. Full Hybrid (Calibrated)	**3**	**0**	**7**	**0**	**100.0%**	**100.0%**	**1.000**

4 Limitations & Conclusion

While deterministic span verification prevents fabricated events from entering the audit graph, overall system recall remains bounded by upstream extraction completeness. Nevertheless, decoupling text extraction from neuro-symbolic protocol and graph verification provides a reproducible, high-specificity instrument for clinical AI governance.

References

[1] M. Gulati *et al.*, "2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain," *Circulation*, vol. 144, no. 22, pp. e368–e454, 2021.

[2] T. H. Kung *et al.*, "Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models," *PLOS Digital Health*, vol. 2, no. 2, p. e0000198, 2023.

[3] L. Zheng *et al.*, "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.